

● MEDIUM

● ALGEBRA

Algebra

10 questions · ~15 min

05

Q1

ALGEBRA

$$2x + y = 37$$

In triangle QRS , sides QR and RS each have a length of x centimeters and side SQ has a length of y centimeters. The given equation represents this situation. Which of the following is the best interpretation of 37 in this context?

- A. The difference, in centimeters, between the lengths of sides QR and SQ
- B. The difference, in centimeters, between the lengths of sides QR and RS
- C. The sum of the lengths, in centimeters, of the three sides of the triangle
- D. The length, in centimeters, of one of the two sides of equal length

The width of a rectangular dance floor is w feet. The length of the floor is 6 feet longer than its width. Which of the following expresses the perimeter, in feet, of the dance floor in terms of w ?

- A. $2w + 6$
- B. $4w + 12$
- C. $w^2 + 6$
- D. $w^2 + 6w$

$$j(x) = mx + 144$$

For the linear function j , m is a constant and $j(12) = 18$. What is the value of $j(10)$?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN ALGEBRA

$$y = \frac{1}{3}x - 14$$

$$y = -x + 18$$

The solution to the given system of equations is x, y . What is the value of x ?

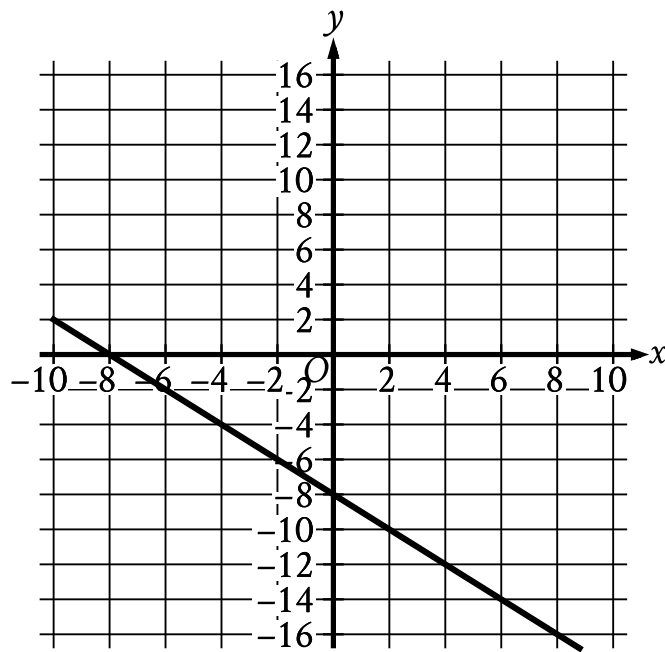
STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A truck can haul a maximum weight of 5,630 pounds. During one trip, the truck will be used to haul a 190-pound piece of equipment as well as several crates. Some of these crates weigh 25 pounds each and the others weigh 62 pounds each. Which inequality represents the possible combinations of the number of 25-pound crates, x , and the number of 62-pound crates, y , the truck can haul during one trip if only the piece of equipment and the crates are being hauled?

- A. $25x + 62y \leq 5,440$
- B. $25x + 62y \geq 5,440$
- C. $62x + 25y \leq 5,630$
- D. $62x + 25y \geq 5,630$



- The line slants gradually down from left to right.
- The line passes through the following points:
 - (negative 8 comma 0)
 - (0 comma negative 8)

What is an equation of the graph shown?

- A. $y = -2x - 8$
- B. $y = x - 8$
- C. $y = -x - 8$
- D. $y = 2x - 8$

Q7

ALGEBRA

$$4x + 12 = \frac{ax + 3}{2}$$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a ?

- A. 0
- B. 3
- C. 8
- D. 12

Q8

ALGEBRA

$$f(x) = \frac{(x + 7)}{4}$$

For the function f defined above, what is the value of $f(9) - f(1)$?

- A. 1
- B. 2
- C. $\frac{1}{4}$
- D. $\frac{9}{4}$

A wire with a length of 106 inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is 6 more than 4 times the value of y . What is the value of x ?

- A. 25
- B. 28
- C. 56
- D. 86

$$y > 4x + 8$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
2	19
4	30
6	41

B.

x	y
2	8
4	16
6	24

C.

x	y
2	13
4	18
6	23

D.

x	y
2	13
4	21
6	29